

## SEMESTER S5

### MICROCONTROLLERS

(Common to CS/CC)

|                                          |                 |                    |                |
|------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                       | <b>PBCST504</b> | <b>CIE Marks</b>   | 60             |
| <b>Teaching Hours/Week<br/>(L:T:P:R)</b> | 3:0:0:1         | <b>ESE Marks</b>   | 40             |
| <b>Credits</b>                           | 4               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>            | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To introduce the ARM architecture and ARM-based microcontroller architecture.
2. To impart knowledge on the hardware and software components to develop embedded systems using STM32 microcontrollers.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Contact Hours</b> |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Introduction to ARM Cortex-M Architecture:-</b><br><br>Overview of Embedded Systems, Applications of Embedded Systems, Introduction to Embedded C, Microcontrollers vs. Microprocessors, Classification of processors, Overview of ARM Cortex-M Series, Introduction to the Cortex-M23 and Cortex-M33 processors and the Armv8-mArchitecture, ARM Core Features: Registers, Memory, and Bus Architecture, Comparison with previous generations of Cortex-M processors.                                                                                                                   | <b>9</b>             |
| <b>2</b>          | <b>STM32 Microcontroller Overview and Peripheral Programming:-</b><br><br>Introduction to STM32 Family, STM32U575 Features and Specifications, Power Management and Low-Power Features Libraries, Introduction to Integrated Development Environment and HAL, Writing, and Debugging Your First Program(LED Interfacing), Interfacing Seven-Segment Display, LCD Display, and Matrix Keypad, Relay Interfacing, Analog to Digital Conversion: Potentiometer, temperature sensor, LDR, Microphone, Digital to Analog Conversion: Simple DAC Output Generation, Generating a Sine Wave, Audio | <b>11</b>            |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          | Signal Generation, Interrupt Handling, Timer and Counter Applications: Basic Timer Configuration, Timers as Counters, Timer-Based Real-Time Clock (RTC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |
| <b>3</b> | <b>Communication Protocols and USB:-</b><br><br>Serial port terminal Application, Serial communication (USART, I2C, SPI, CAN), Interfacing an I2C Temperature Sensor and Displaying Data on an LCD, writing to and Reading from an SPI-based EEPROM, Configuring and Implementing CAN Communication between Multiple STM32U575 Microcontrollers, Creating a USB HID Device for Keyboard / Mouse Emulation                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>10</b> |
| <b>4</b> | <b>IoT, Wireless Communication, and RTOS:-</b><br><br>Introduction to IoT, IoT Architecture, Protocols (MQTT, CoAP), IoT Security Principles and Common Threats Wireless Communication: Interfacing GSM (Call, SMS, Internet), Bluetooth Communication Basics, LoRa Communication Basics and Applications, Designing an IoT-Based Home Automation System, Introduction to RTOS Concepts, FreeRTOS with STM32: Task Creation, Scheduling, and Management, RTOS Timers, Delays, and RTC Integration, Inter-task Communication: Queues and Semaphores<br><br><b>Trust Zone Technology:</b> Introduction to ARM Trust Zone, Trust Zone Architecture and Features, Secure and Non-Secure Worlds: Configuration and Management, Implementing Trust Zone in STM32U575, Advanced Debugging and Optimization: Code and Memory Optimization Techniques, Debugging Strategies and Tools | <b>14</b> |

#### **Suggestion on Project Topics**

- Identify real world problems requiring hardware solutions and develop them using peripheral devices. Some of the examples would be - Home automation, Small home/office security system, ARM based voice response system etc.

**Course Assessment Method**  
**(CIE: 60 marks, ESE: 40 marks)**

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Project | Internal Ex-1 | Internal Ex-2 | Total |
|------------|---------|---------------|---------------|-------|
| 5          | 30      | 12.5          | 12.5          | 60    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                   | Part B                                                                                                                                                                                                    | Total     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"> <li>2 Questions from each module.</li> <li>Total of 8 Questions, each carrying 2 marks<br/>(8x2 =16 marks)</li> </ul> | 2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 subdivisions. Each question carries 6 marks.<br><br><b>(4x6 = 24 marks)</b> | <b>40</b> |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                                  | Bloom's Knowledge Level (KL) |
|----------------|--------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Explain the architectural features and instructions of the ARM microcontrollers.                 | <b>K2</b>                    |
| <b>CO2</b>     | Develop applications involving interfacing of external devices and I/O with ARM microcontroller. | <b>K3</b>                    |
| <b>CO3</b>     | Use various communication protocols of interaction with peer devices and peripherals.            | <b>K3</b>                    |
| <b>CO4</b>     | Demonstrate the use of a real time operating system in embedded system applications.             | <b>K3</b>                    |
| <b>CO5</b>     | Apply hardware security features of ARM in real world applications.                              | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table:**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO2</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO3</b> | 3   | 3   | 3   | 3   | 3   |     |     |     |     |      |      | 3    |
| <b>CO4</b> | 3   | 3   | 3   | 3   | 3   |     |     |     |     |      |      | 3    |
| <b>CO5</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 3    |

| <b>Text Books</b> |                                                                |                             |                              |                         |
|-------------------|----------------------------------------------------------------|-----------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                                       | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors | Joseph Yiu                  | Newnes - Elsevier            | 3/e, 2014               |
| 2                 | Mastering STM32                                                | Carmine Noviello            | Learnpub                     | 2/e, 2022               |

| <b>Reference Books</b> |                                                           |                                                         |                              |                         |
|------------------------|-----------------------------------------------------------|---------------------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                                  | <b>Name of the Author/s</b>                             | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | ARM System Developer's Guide                              | Andrew N. Sloss, Dominic Symes, Chris Wright            | Morgan Kaufman               | 1/e, 2008               |
| 2                      | Embedded System Design with Arm Cortex-M Microcontrollers | Cem Ünsalan, Hüseyin Deniz Gürhan<br>Mehmet Erkin Yücel | Springer                     | 1/e, 2022               |
| 3                      | Introduction to ARM® Cortex-M Microcontrollers            | Jonathan W. Valvano                                     | Self-Published               | 5/e, 2014               |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                     |
| 1                              | <a href="https://archive.nptel.ac.in/courses/106/105/106105193/">https://archive.nptel.ac.in/courses/106/105/106105193/</a> |
| 2                              | <a href="https://www.st.com/resource/en/datasheet/">https://www.st.com/resource/en/datasheet/</a>                           |

### PBL Course Elements

| L: Lecture<br>(3 Hrs.)                                    | R: Project (1 Hr.), 2 Faculty Members    |                                              |                                                                                                            |
|-----------------------------------------------------------|------------------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------|
|                                                           | Tutorial                                 | Practical                                    | Presentation                                                                                               |
| Lecture delivery                                          | Project identification                   | Simulation/<br>Laboratory Work/<br>Workshops | Presentation<br>(Progress and Final<br>Presentations)                                                      |
| Group discussion                                          | Project Analysis                         | Data Collection                              | Evaluation                                                                                                 |
| Question answer<br>Sessions/<br>Brainstorming<br>Sessions | Analytical thinking and<br>self-learning | Testing                                      | Project Milestone Reviews,<br>Feedback,<br>Project reformation (If<br>required)                            |
| Guest Speakers<br>(Industry<br>Experts)                   | Case Study/ Field<br>Survey Report       | Prototyping                                  | Poster Presentation/<br>Video Presentation: Students<br>present their results in a 2 to 5<br>minutes video |

## **Assessment and Evaluation for Project Activity**

| <b>Sl. No</b> | <b>Evaluation for</b>                                               | <b>Allotted Marks</b> |
|---------------|---------------------------------------------------------------------|-----------------------|
| 1             | Project Planning and Proposal                                       | 5                     |
| 2             | Contribution in Progress Presentations and Question Answer Sessions | 4                     |
| 3             | Involvement in the project work and Team Work                       | 3                     |
| 4             | Execution and Implementation                                        | 10                    |
| 5             | Final Presentations                                                 | 5                     |
| 6             | Project Quality, Innovation and Creativity                          | 3                     |
| <b>Total</b>  |                                                                     | <b>30</b>             |

### **1. Project Planning and Proposal (5 Marks)**

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

### **2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

### **3. Involvement in the Project Work and Team Work (3 Marks)**

- Active participation and individual contribution
- Teamwork and collaboration

### **4. Execution and Implementation (10 Marks)**

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

**5. Final Presentation (5 Marks)**

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

**6. Project Quality, Innovation, and Creativity (3 Marks)**

- Overall quality and technical excellence of the project
- Innovation and originality in the project

Creativity in solutions and approaches